FLOOD WATCH

By Mag4

MITIGATING URBAN FLOOD RISKS THROUGH SMARTER SYSTEM

KOH EE HUI JOSEPH (1441309)
KHIN KAN KAW LIN (1485064)
MAY YU YA MAW (8950362)
SUZAINA SADIMIN (1037195)
VINCENT (115005)

Agenda

01 Introduction

02 Overall Architecture

03 Key Functions

04 AI & Advanced Tech

05 Security

06 Conclusion

What is FLOODWATCH?

Smart, Proactive Enhancement to Singapore's Existing Flood Management System

Singapore's Existing Approach	VS	FloodWatch Enhancement
Radar + Sensor Monitoring Forecast rainfall & monitor drainage conditions to anticipate potential flood locations		**AI Flood Prediction** Combine multi-modal data to predict flood risk and infrastructure impact
Periodic & Manual Inspection Identify infrastructure issues through inspections and monitoring		**Continuous AI Monitoring** Detect potential pump failures and drainage blockages earlier
Public Alert Broadcast Sends flood warnings through multiple channels		**Smart Alert Escalation** Tracks acknowledgement and escalates critical alerts when necessary
Localised Storage Management Manages stormwater storage based mainly on local water levels and infrastructure conditions		**AI-Driven Storage Optimisation** Coordinates available storage across multiple locations based on predicted flood conditions
Segmented Dispatch & Manual Routing Field units rely on standalone alerts & manual coordination, risking navigation delays through flooded sectors		**Tactical Situational Awareness Feeds** Converts live data into real-time operational coordination feeds, routing emergency units dynamically to flooded areas

System Goals

FLOOD WATCH

AI-Based Flood Prediction & Risk Analysis

- **Multi-Modal Ingestion**
- **Predictive AI Models** to forecast street-level risks
- **Impact Overlay Mapping**: Pinpoints exact roads & buildings inside the hazard zone

1

AI-Powered Drainage & Pump Health Monitoring

- Continuous Machine Telemetry
- Computer Vision Blockage Detection
- Predictive Asset Maintenance

2

Early Warning & Multi-Channel Alert Dissemination

- **Serverless Fan-Out**: Delivers rapid, concurrent alerts across multiple agency channels
- **Response Accountability**: Tracks live delivery & responder message acknowledgements
- **Automated Ticket Escalation**

3

Stormwater Storage Optimisation

- 3-Phase Proactive Preparation
  - Pre-Storm Flushing
  - Dynamic Surface Diversion
  - Graph-Based Staggered Recovery

4

Tactical Ground Situational Awareness & Operational Feeds

- **Emergency Routing:** Finds flood -free routes for rescue teams
- **Dynamic Traffic Control:** Identifies roads requiring closure/diversion
- **Field Intelligence:** Consolidates geo-tagged photos &reports for situational awareness

5

National Environment Agency Safeguard • Nurture • Cherish

Land Transport Authority

PUB SINGAPORE'S NATIONAL WATER AGENCY

SCDF The Life Saving Force ...for a safer Singapore

Meteorological Service Singapore

Overall Architecture

01

Data Sources

NEA/MSS Weather API, Water/Flow/Rain Sensors, Pump & Vibration Sensors, CCTV Feeds, Public Reports

02

Network & Ingestion

NB-IoT/LTE-M (static sensors), 5G/Fibre (videos, actuators), AWS IoT Core, Amazon Kinesis

03

Processing & Intelligence

AWS Lambda, AWS IoT Events (CEP), SageMaker (LSTM, RandomForest), SageMaker (ST-GNN, YOLOv11)

04

Multi-Modal Storage

Timestream (time-series), PostGIS (spatial), DynamoDB (document), Neptune (graph), QLDB (ledger)

05

Application Layer

FloodWatch Dashboard, Digital Twin (IoT TwinMaker), Warning Decision Layer

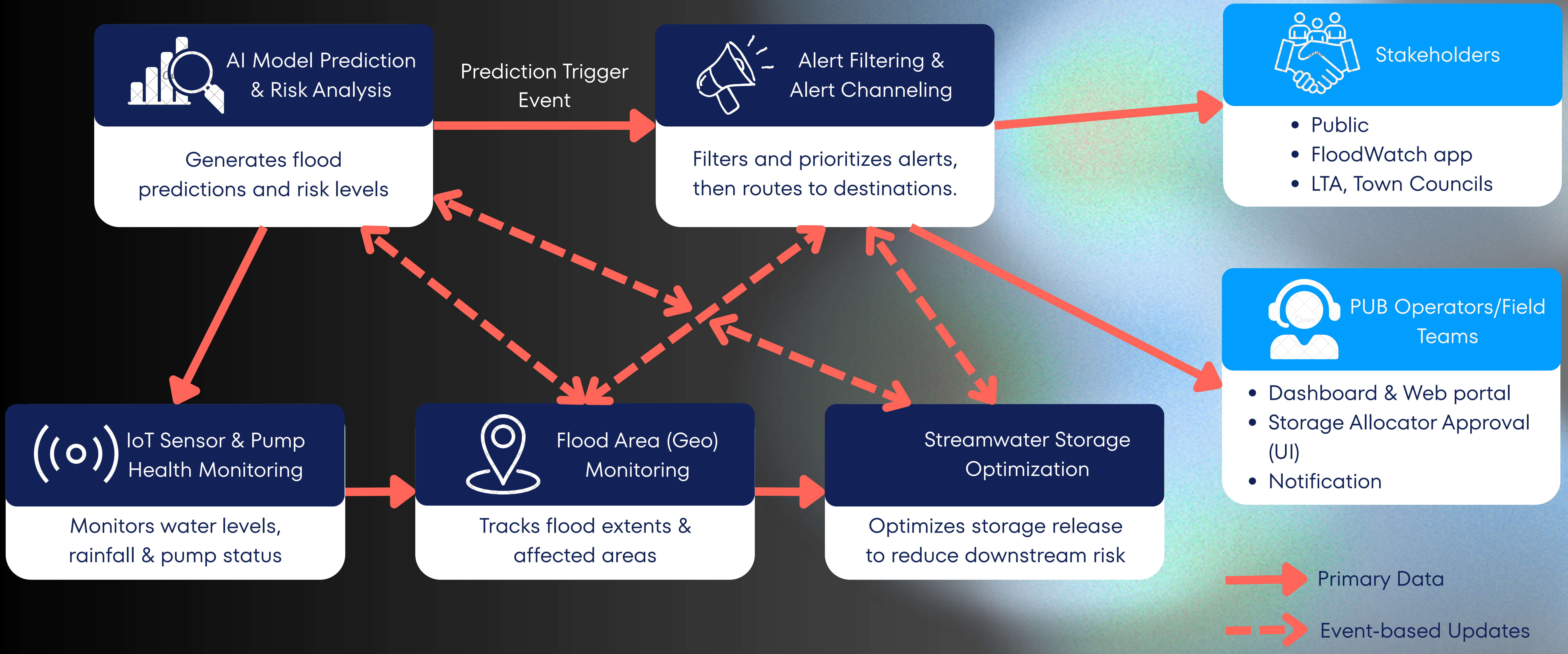
06

Dissemination & Stakeholders

Cell Broadcast + App (Public), PUB/SCDF/SPF/LTA/Town Councils

HOW THE 5 FUNCTIONS CONNECT?

One Integrated system, not 5 independent tools



Key Functions I

DRAINAGE & PUMP HEALTH MONITORING

FLOOD WATCH

OBJECTIVE:

TO CONTINUOUSLY MONITOR GRAINAGE AND PUMP HEALTH AND AUTOMATE WORK ORDER FOR MAINTENANCE

INFRASTRUCTURE HEALTH

Asset	Vibration	Motor Temp	Health Score	Status
P001	2.1 mm/s	48°C	92	Low
P014	5.8 mm/s	61°C	68	Moderate
P022	9.4 mm/s	79°C	31	High

HEALTH SCORING

AWS IOT SENSORS
AWS LAMBDA
THRESHOLD RULES

DIGITAL TWIN

AWS IOT TWINMAKER
GIS MAPPING DATA

WORK ORDER

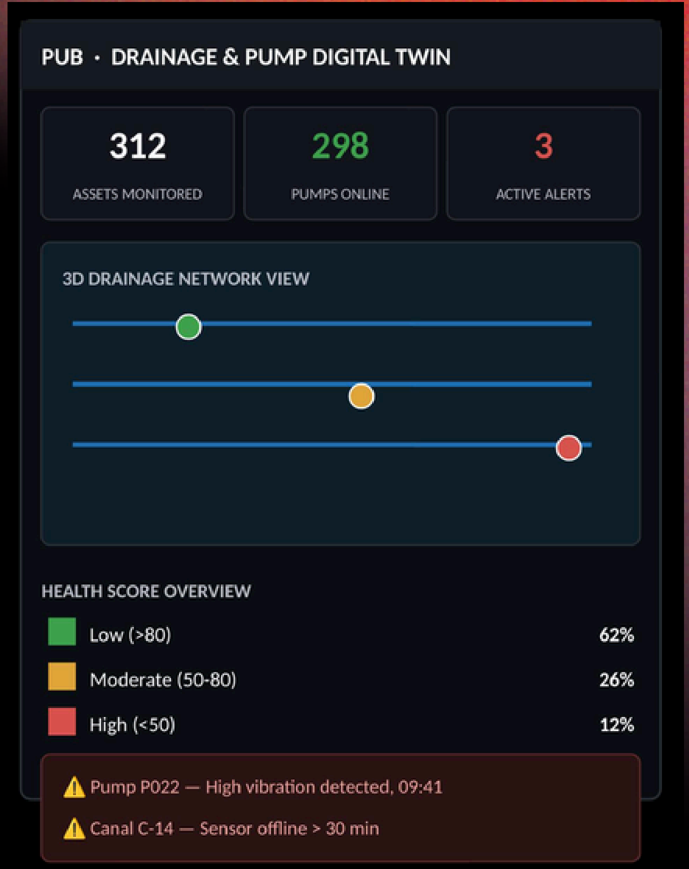
AUTOMATIC TRIGGER
NO MANUAL INSPECTION

WORK ORDER

WARNING

assest p02 - Bearing Wear
recommended : Replace Bearing
Status : Pending

VISUALIZATION DASHBOARD



Key Functions II

FLOOD WATCH

STORMWATER STORAGE OPTIMISATION

OBJECTIVE:
COORDINATE STORMWATER STORAGE ASSETS BEFORE, DURING & AFTER RAINFALL TO MAXIMISE AVAILABLE CAPACITY

STORAGE CAPACITY OVERVIEW

Asset	Current Level	Available	Recommended Action	Status
Detention Tank 01	62%	1,200 m³	Divert Inflow	Optimal
Canal 05	81%	340 m³	Monitor	Moderate
Storage Site 12	95%	40 m³	Activate Overflow	Critical

AI OPTIMIZATION

AWS SAGEMAKER
ST-GNN MODEL
YOLOV11

DIGITAL TWIN

AWS IOT TWINMAKER
STORAGE NETWORK SIM

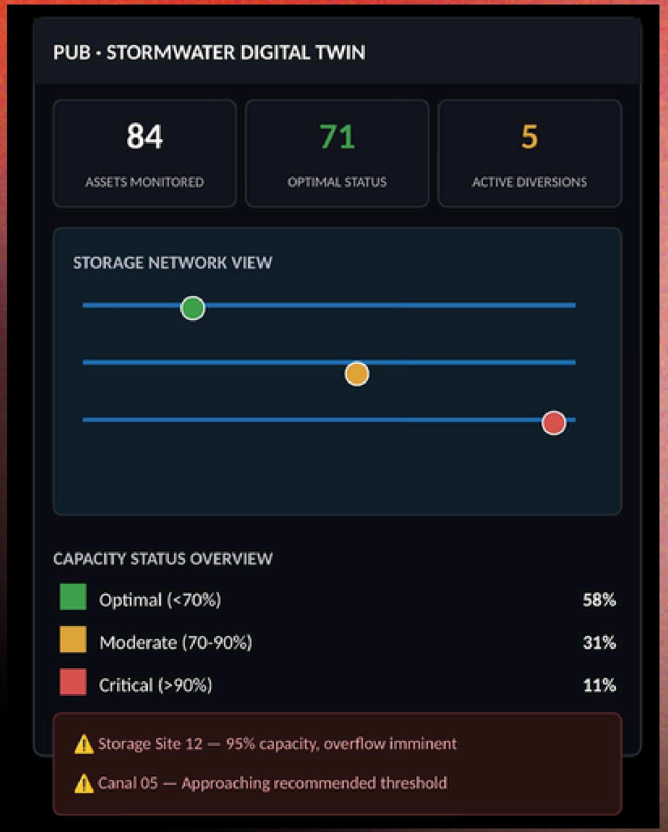
SAFETY VERIFICATION

AMAZON NEPTUNE
POSTGRESQL + POSTGIS

OPTIMISATION REPORT

Storage_Site_12 · 95% Full
Recommended: Activate Overflow
Status: Pending Approval

VISUALIZATION BOARD



Key Functions III
TACTICAL GROUND SITUATIONAL AWARENESS & OPERATIONAL FEEDS

FLOOD WATCH

OBJECTIVE:
TO TRACK AREAS AFFECTED BY FLASH FLOODS IN REAL TIME, REROUTE AND ALERT

SECTOR STATUS

Sector	Road Status	Water Level	Dispatched To	Alert
Toa Payoh	Clear	0.1 m	—	Low
Bedok	Slow	0.3 m	LTA / Traffic Police	Moderate
Geylang	Blocked	0.6 m	SCDF + PUB QRT	High

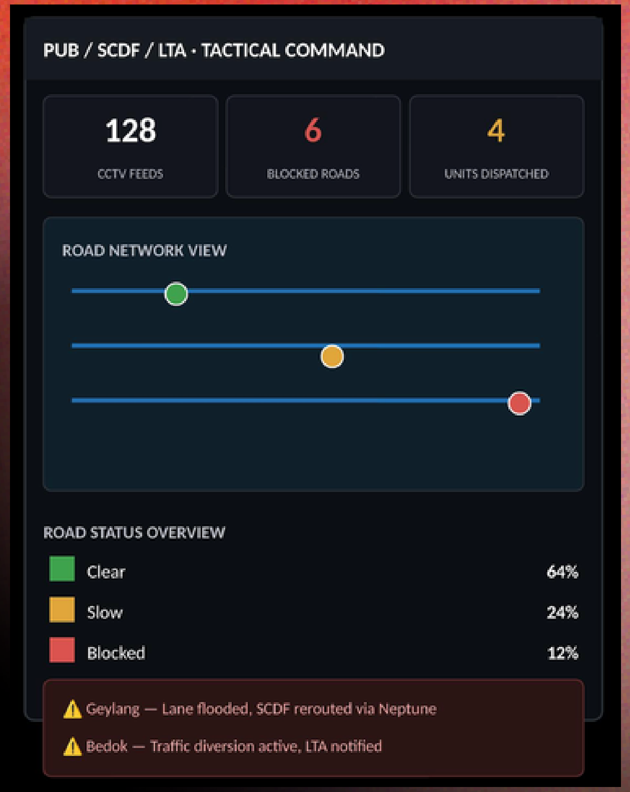
ROAD DETECTION
YOLOV11
SAGEMAKER ENDPOINT
CCTV FRAME ANALYSIS

AUTO ESCALATION
AMAZON EVENTBRIDGE
TRIGGERS ALERTS

TACTICAL ROUTING
AMAZON NEPTUNE
OPENCYPHER
SHORTEST-PATH

INCIDENT REPORT

Geylang · Lane Flooded
Photo: crowdsourced (verified)
Status: SCDF Dispatched



Key Functions IV

AI-BASED FLOOD PREDICTION & RISK ANALYSIS

OBJECTIVE:
TO PREDICT FLOODING AND PERFORM IMPACT ANALYSIS

FLOOD PROBABILITY

PREDICTED FLOOD RISK OVERVIEW

Current Water Level	Predicted Water Level (LSTM + Random Forest)	Rainfall	Duration	Flood Risk
1.4 m	1.8 m	5 mm	Short	Low
1.4 m	2.6 m	25 mm	Short	Moderate
1.4 m	2.9 m	40 mm	Long	High

AI PREDICTION

AWS SAGEMAKER
AI MODELS:
LTSM
RANDOM FOREST

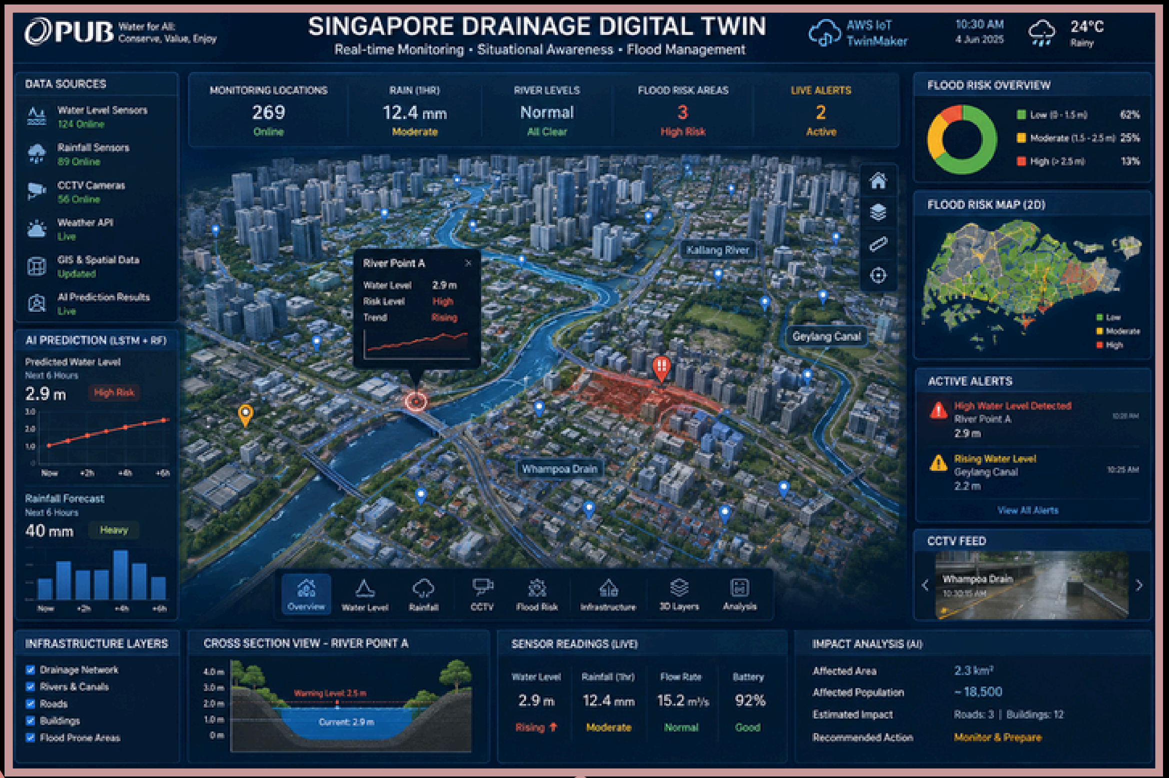
DIGITAL TWIN

AWS IOT TWINMAKER
GIS MAPPING DATA

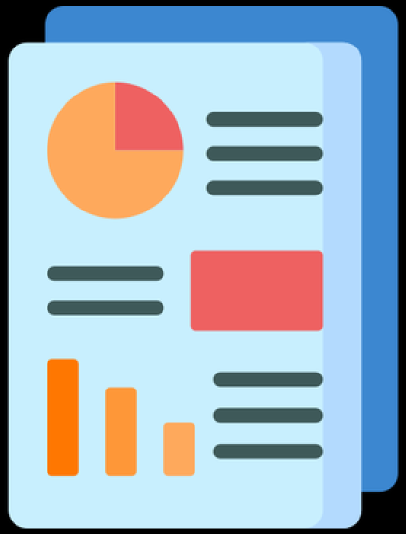
INCIDENT REPORT

AWS LAMBDA
AWS DYNAMO DB
JSON-STYLE FORMAT

VISUALIZATION DASHBOARD



DOCUMENT



Key Functions V

EARLY WARNING & MULTI-CHANNEL ALERT DISSEMINATION

OBJECTIVE:
SEND OUT NOTIFICATIONS TO ALERT THE PUBLIC OF FLOODING EVENTS

TYPES OF ALERTS

MULTI-CHANNEL ALERTS

SMS
FLOODWATCH API
ROAD SIGNAGES

ALERT LOGGING & AUDIT TRAIL

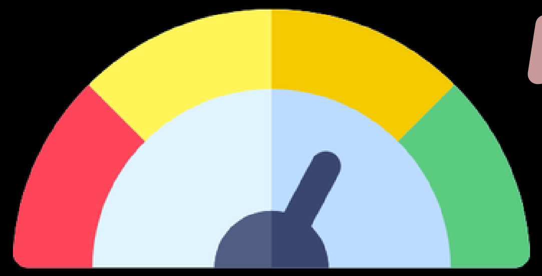
AWS LAMBDA
AWS DYNAMO DB

LOGS

SEVERITY SCORING

AWS SAGEMAKER
AI MODELS:
LTSM
RANDOM FOREST

FLOOD SEVERITY



Advanced Tech & AI in FloodWatch

AWS IoT TwinMaker

- **3D virtual replica** of Singapore's drainage map, roads and traffic infrastructure.
- **Real-Time** updates from the sensors & GIS spatial mapping.
- **Simulation** allows relevant authorities like PUB to perform simulations before it happens.

Edge Computer

- **Real-Time** CCTV camera feeds uses edge computing to detect objects.
- **Identify** alternative stormwater storages which are safe to use.
- **YOLOv11** processes live CCTV feeds to find out the objects interfering with the drainage causing the floods.

Spatio-Temporal Graph (AI)

- **Topology** shows the water linkages using graph nodes (canals) & edges (water channels) .
- **Optimizes** its AI to provide the most ideal way to route the flood to the canals.
- **Pairs** with LTSM water level prediction to create diversion paths.

NB-IoT + 5G Next-Gen

- **NB-IoT** system provides a long battery life with strong signal which supports thousands of devices without network overload.
- **5G Next-Gen** allows a high speed internet for the live CCTV feeds to be displayed.

Security in FloodWatch

MQTT End-to-End Encryption

Using MQTT EE2E enhances security in our system as it does not allow third parties to read the application data and remained the sensitive fields protected.

X.509 Authentication

X.509 certificate encrypts data and verify signatures to allow for safe and encrypted communication.

Amazon QLDB

Amazon QLDB allows for transaction logs to be kept un-tampered. Once the logs are stored and kept, it cannot be overwritten or erased even by the administrator.

Conclusion

Predictive Analysis

- FloodWatch uses AI to predict possible outcomes.
- The prediction generates early warning to the relevant authorities to be prepared and anticipate possible emergencies.

Multi-Agency Data

- FloodWatch obtains data from reputable government data sources. (e.g. LTA, SCDF, PUB, LTA, Town Councils)
- Synchronizes all the collected data and analyse it for predictions.

Widespread Reach

- FloodWatch disseminate messages to all members of the public via all Telco Cell Broadcast for a guaranteed reach.
- Emergency messages does not require app opt-in for public members to be notified.

QUESTIONS?

THANK YOU!